

Refined 3D Index — Full Report

m003

Generated by manifold-index

March 30, 2026

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1 Manifold Overview

Manifold name: **m003**

- Number of ideal tetrahedra: $n = 2$
- Number of cusps: $r = 1$
- Internal edges: $n - r = 1$ (easy: 0, hard: 1)
- Computation order: $q^{20/2}$
- Boundary curve range: $m \in \{-2, -1, 0, 1, 2\}$, $e \in \{-2, -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$

2 Ideal Triangulation (SnaPy Data)

Gluing equation matrix from SnaPy, shape (4×6) .

Edge	(Z_1, Z'_1, Z''_1)	(Z_2, Z'_2, Z''_2)
0	$(2, 0, 1)$	$(2, 0, 1)$
1	$(0, 2, 1)$	$(0, 2, 1)$

Cusp	(Z_1, Z'_1, Z''_1)	(Z_2, Z'_2, Z''_2)
α	$(0, -2, 0)$	$(2, 0, 0)$
β	$(0, -1, 0)$	$(2, -1, 0)$

3 Edge Classification

Easy edges: 0 total, 0 independent. Hard edges: 1.

Hard Edges		
Index	Symbolic Equation	Triplets (f_i, g_i, h_i)
H_0	$2Z_1 + Z''_1 + 2Z_2 + Z''_2 = 2$	$(2, 0, 1) (2, 0, 1)$

4 Neumann–Zagier Data

$g_{\text{NZ}} \in \text{Sp}(2n, \mathbb{Q})$, 4×4 , column ordering $(Z_1, \dots, Z_n, Z''_1, \dots, Z''_n)$.

$$g_{\text{NZ}} = \begin{pmatrix} 2 & 2 & 2 & 0 \\ 2 & 2 & 1 & 1 \\ \frac{1}{2} & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Row labels (right margin): $\mu_0, H_0, \lambda_0/2, \Gamma_0$.

Affine shifts: $\nu_x = (-2, -2)$, $\nu_p = (-1, 0)$.

5 3D Index Formula

The 3D index is computed via:

$$I(\mathbf{m}_{\text{ext}}, \mathbf{e}_{\text{ext}}) = \sum_{\mathbf{e}_{\text{int}} \in (\frac{1}{2}\mathbb{Z})^{n-r}} (-q^{1/2})^{\mathbf{m} \cdot \nu_p - \mathbf{e} \cdot \nu_x} \prod_{a=1}^n I_{\Delta}(m_a, e_a)$$

where $\kappa = (\mathbf{m}_{\text{ext}}, \mathbf{0}^{n-r}, \mathbf{e}_{\text{ext}}, \mathbf{e}_{\text{int}})$ and the local charges are:

$$m_a = (g_{\text{NZ}}^{-1} \kappa)_a, \quad e_a = (g_{\text{NZ}}^{-1} \kappa)_{n+a}$$

Only terms with integer (m_a, e_a) contribute.

Explicit Local Charges. Δ_1 : $m_1 = \frac{1}{2}m_0 - 2e_0 - e_{\text{int},0}$, $e_1 = -\frac{1}{2}m_0 + 2e_0 + 2e_{\text{int},0}$

Δ_2 : $m_2 = \frac{1}{2}m_0 - e_{\text{int},0}$, $e_2 = -\frac{3}{2}m_0 + 2e_0 + 2e_{\text{int},0}$

The tetrahedron index $I_{\Delta}(m, e)$ is given by

$$I_t(m, e) = \sum_{k=\max(0, -e)}^{\infty} \frac{(-1)^k q^{k(k+1)/2 - (2k+e)m/2}}{\prod_{j=1}^k (1 - q^j) \prod_{j=1}^{k+e} (1 - q^j)}$$

with $I_{\Delta}(m, e) = (-q^{1/2})^m \cdot I_t(-m - e, m)$ when $m + e \geq 0$, else $I_{\Delta}(m, e) = I_t(m, e)$.

6 Refined Index Results

Series truncated at $q^{20/2}$ (i.e. $N_{\text{max}} = 10$). Number of computed sectors: 45. Non-zero sectors: 27.

$$\begin{aligned}
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-2)) &= q^2 + q^3 + q^3 \eta_0 + q^4 + 2q^4 \eta_0 + 3q^5 \eta_0 - 2q^6 + 2q^6 \eta_0 - 6q^7 - q^7 \eta_0 - q^7 \eta_0^2 - 11q^8 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-3/2)) &= -q^{1/2} + q^{5/2} + 3q^{7/2} + q^{7/2} \eta_0 + 5q^{9/2} + 2q^{9/2} \eta_0 - q^{11/2} \eta_0^{-1} + 5q^{11/2} + 3q^{11/2} \eta_0 - \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-1)) &= -q - q^2 + q^3 \eta_0^{-1} + q^3 + 2q^4 \eta_0^{-1} + 4q^4 + q^4 \eta_0 + 2q^5 \eta_0^{-1} + 8q^5 + q^5 \eta_0 + q^6 \eta_0^{-1} + \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (-1/2)) &= -q^{3/2} \eta_0^{-1} - q^{3/2} + 2q^{7/2} \eta_0^{-1} + 2q^{7/2} + 5q^{9/2} \eta_0^{-1} + 5q^{9/2} + 7q^{11/2} \eta_0^{-1} + 7q^{11/2} \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (0)) &= -q \eta_0^{-1} - q^2 \eta_0^{-1} + q^3 \eta_0^{-1} + q^3 + q^4 \eta_0^{-2} + 4q^4 \eta_0^{-1} + 2q^4 + q^5 \eta_0^{-2} + 8q^5 \eta_0^{-1} + 2q^5 \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (1/2)) &= -q^{1/2} \eta_0^{-1} + q^{5/2} \eta_0^{-1} + q^{7/2} \eta_0^{-2} + 3q^{7/2} \eta_0^{-1} + 2q^{9/2} \eta_0^{-2} + 5q^{9/2} \eta_0^{-1} + 3q^{11/2} \eta_0^{-2} + \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (1)) &= q^2 \eta_0^{-1} + q^3 \eta_0^{-2} + q^3 \eta_0^{-1} + 2q^4 \eta_0^{-2} + q^4 \eta_0^{-1} + 3q^5 \eta_0^{-2} + 2q^6 \eta_0^{-2} - 2q^6 \eta_0^{-1} - q^7 \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (3/2)) &= q^{5/2} \eta_0^{-2} + q^{7/2} \eta_0^{-2} + q^{9/2} \eta_0^{-2} - q^{9/2} \eta_0^{-1} - q^{11/2} \eta_0^{-1} - q^{13/2} \eta_0^{-3} - 3q^{13/2} \eta_0^{-2} - 2q^{13/2} \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (-2), \mathbf{e} = (2)) &= -q^5 \eta_0^{-2} - q^6 \eta_0^{-3} - 2q^6 \eta_0^{-2} - 2q^7 \eta_0^{-3} - 3q^7 \eta_0^{-2} - 4q^8 \eta_0^{-3} - 4q^8 \eta_0^{-2} + q^8 \eta_0^{-1} - 6 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (-1), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-2)) &= q^3 \eta_0^2 + q^4 \eta_0^2 + 2q^5 \eta_0^2 - q^6 \eta_0 - q^6 \eta_0^3 - 2q^7 \eta_0 - 2q^7 \eta_0^2 - 2q^7 \eta_0^3 - 4q^8 \eta_0 - 8q^8 \eta_0^2 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-3/2)) &= q^3 \eta_0 + q^3 \eta_0^2 + 2q^4 \eta_0 + 2q^4 \eta_0^2 + 3q^5 \eta_0 + 3q^5 \eta_0^2 - q^6 + 2q^6 \eta_0 + 2q^6 \eta_0^2 - q^6 \eta_0^3 - q^7 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-1)) &= -q \eta_0 - q^2 \eta_0 + q^3 + q^3 \eta_0 + q^3 \eta_0^2 + q^4 + 4q^4 \eta_0 + q^4 \eta_0^2 + 2q^5 + 8q^5 \eta_0 + 2q^5 \eta_0^2 + 9 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (-1/2)) &= -q - q \eta_0 - q^2 - q^2 \eta_0 + q^3 + q^3 \eta_0 + 4q^4 + 4q^4 \eta_0 + 8q^5 + 8q^5 \eta_0 - q^6 \eta_0^{-1} + 9q^6 + \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (0)) &= 1 - 2q - 3q^2 + q^3 \eta_0^{-1} + q^3 \eta_0 + 2q^4 \eta_0^{-1} + 4q^4 + 2q^4 \eta_0 + 3q^5 \eta_0^{-1} + 12q^5 + 3q^5 \eta_0 \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (1/2)) &= -q \eta_0^{-1} - q - q^2 \eta_0^{-1} - q^2 + q^3 \eta_0^{-1} + q^3 + 4q^4 \eta_0^{-1} + 4q^4 + 8q^5 \eta_0^{-1} + 8q^5 - q^6 \eta_0^{-2} - \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (1)) &= -q \eta_0^{-1} - q^2 \eta_0^{-1} + q^3 \eta_0^{-2} + q^3 \eta_0^{-1} + q^3 + q^4 \eta_0^{-2} + 4q^4 \eta_0^{-1} + q^4 + 2q^5 \eta_0^{-2} + 8q^5 \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (3/2)) &= q^3 \eta_0^{-2} + q^3 \eta_0^{-1} + 2q^4 \eta_0^{-2} + 2q^4 \eta_0^{-1} + 3q^5 \eta_0^{-2} + 3q^5 \eta_0^{-1} - q^6 \eta_0^{-3} + 2q^6 \eta_0^{-2} + 2q^6 \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (0), \mathbf{e} = (2)) &= q^3 \eta_0^{-2} + q^4 \eta_0^{-2} + 2q^5 \eta_0^{-2} - q^6 \eta_0^{-3} - q^6 \eta_0^{-1} - 2q^7 \eta_0^{-3} - 2q^7 \eta_0^{-2} - 2q^7 \eta_0^{-1} - 4q^8 \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (-1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (0)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (1/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (1)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (3/2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (1), \mathbf{e} = (2)) &= 0 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-2)) &= -q^5 \eta_0^4 - q^6 \eta_0^3 - 2q^6 \eta_0^4 - 2q^7 \eta_0^3 - 3q^7 \eta_0^4 - 4q^8 \eta_0^3 - 4q^8 \eta_0^4 + q^8 \eta_0^5 - 6q^9 \eta_0^3 - 4q^9 \eta_0^4 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-3/2)) &= q^{5/2} \eta_0^3 + q^{7/2} \eta_0^3 + q^{9/2} \eta_0^3 - q^{9/2} \eta_0^4 - q^{11/2} \eta_0^4 - q^{13/2} \eta_0^2 - 3q^{13/2} \eta_0^3 - 2q^{13/2} \eta_0^4 - 2 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-1)) &= q^2 \eta_0^3 + q^3 \eta_0^2 + q^3 \eta_0^3 + 2q^4 \eta_0^2 + q^4 \eta_0^3 + 3q^5 \eta_0^2 + 2q^6 \eta_0^2 - 2q^6 \eta_0^3 - q^7 \eta_0 - q^7 \eta_0^2 - 6 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (-1/2)) &= -q^{1/2} \eta_0^2 + q^{5/2} \eta_0^2 + q^{7/2} \eta_0 + 3q^{7/2} \eta_0^2 + 2q^{9/2} \eta_0 + 5q^{9/2} \eta_0^2 + 3q^{11/2} \eta_0 + 5q^{11/2} \eta_0^2 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (0)) &= -q \eta_0 - q^2 \eta_0 + q^3 \eta_0 + q^3 \eta_0^2 + q^4 + 4q^4 \eta_0 + 2q^4 \eta_0^2 + q^5 + 8q^5 \eta_0 + 2q^5 \eta_0^2 + q^6 + 9 \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (1/2)) &= -q^{3/2} - q^{3/2} \eta_0 + 2q^{7/2} + 2q^{7/2} \eta_0 + 5q^{9/2} + 5q^{9/2} \eta_0 + 7q^{11/2} + 7q^{11/2} \eta_0 - q^{13/2} \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (1)) &= -q^{3/2} - q^{3/2} \eta_0 + 2q^{7/2} + 2q^{7/2} \eta_0 + 5q^{9/2} + 5q^{9/2} \eta_0 + 7q^{11/2} + 7q^{11/2} \eta_0 - q^{13/2} \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (3/2)) &= -q^{3/2} - q^{3/2} \eta_0 + 2q^{7/2} + 2q^{7/2} \eta_0 + 5q^{9/2} + 5q^{9/2} \eta_0 + 7q^{11/2} + 7q^{11/2} \eta_0 - q^{13/2} \eta_0^{-1} \\
I^{\text{ref}}(\mathbf{m} = (2), \mathbf{e} = (2)) &= -q^{3/2} - q^{3/2} \eta_0 + 2q^{7/2} + 2q^{7/2} \eta_0 + 5q^{9/2} + 5q^{9/2} \eta_0 + 7q^{11/2} + 7q^{11/2} \eta_0 - q^{13/2} \eta_0^{-1}
\end{aligned}$$

7 Weyl Symmetry

Define the Weyl-shifted index:

$$f(\eta_j; m, e) = \eta_j^{a_j \cdot e + b_j \cdot m} \cdot I^{\text{ref}}(m, e)$$

Weyl symmetry requires $f(m, e) = f(-m, -e)$ for each sector pair, where f denotes the Weyl-shifted index above.

Hard edge j	a_j	b_j	$a_j \in \mathbb{Z}?$	$2b_j \in \mathbb{Z}?$
0	1	$-\frac{1}{2}$	✓	✓

$a \in \mathbb{Z}^k$, $b \in (\frac{1}{2}\mathbb{Z})^k$ — Dehn filling compatible.

Edge compatibility: each hard edge j must have $a_j \in \mathbb{Z}$; incompatible edges are collapsed ($\eta_j = 1$).

H₀: **PASS**

Weyl symmetry $f(m, e) = f(-m, -e)$: 45/45 sectors pass **PASS**

The adjoint $\text{su}(2)$ projection (eq. 2.59–2.61) integrates out η and cusp fugacities against the Haar measure weighted by the adjoint character; the result must equal -1 per filled cusp.

Intermediate c_e values ((q^1, η^0) coefficient of $I^{\text{ref}}(m = 0, e)$):

e	c_e
-2	0
$-\frac{3}{2}$	0
-1	-1
$-\frac{1}{2}$	0
0	-2
$\frac{1}{2}$	0
1	-1
$\frac{3}{2}$	0
2	0

Projected value: -1 (expected -1): **PASS**

8 Dehn Filling

8.1 Filling 1: Cusp 0, slope $(1, 2)$

8.1.1 Slope Transformation

- User slope (physical): $P = 1$, $Q = 2$ (cycle $1 \cdot M + 2 \cdot L$)
- Non-closable cycle: $\gamma = -1 \cdot M + 1 \cdot L$ (slope $-1/1$)
- $\text{SL}(2, \mathbb{Z})$ complement: $\delta = 0 \cdot M + -1 \cdot L$
- Verification: $\det \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix} = 1$ (should be $+1$)
- User slope in (γ, δ) basis: $(p, q) = (-1, -3)$

8.1.2 Hirzebruch–Jung Continued Fraction

$p/q = -1/-3$: $[1, 2, 2]$, $\ell = 3$.

8.1.3 g_{NZ} After Basis Change

After replacing cusp basis to the NC cycle $(-1, 1)$:

$$g'_{\text{NZ}} = \begin{pmatrix} -1 & 1 & -1 & 1 \\ 2 & 2 & 1 & 1 \\ -\frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\nu'_x = (0, -2), \nu'_p = (1, 0).$$

8.1.4 Physical Weyl Vectors

$$\mathbf{a}_{\text{phys}} = (0)$$

$$\mathbf{b}_{\text{phys}} = (\frac{1}{2})$$

All hard edges are compatible ($a_j \in \mathbb{Z}$).

8.1.5 Filled Refined Index

HJ: $[1, 2, 2]$, $\ell = 3$, kernel terms: 383.

$$I_{1/2}^{\text{ref, filled}} = 1 + \eta_0 \eta^{-8V_0} - q - q^2 - q^2 \eta^{4V_0} + q^3 \eta_0^{-1} \eta^{8V_0} - q^3 \eta^{-4V_0} + q^3 - 2q^3 \eta^{8V_0} - 2q^3 \eta_0 \eta^{-8V_0} - q^4 \eta_0^{-1} \eta^{4V_0} + q^4 \eta_0^{-1} \eta^{8V_0} +$$

8.2 Filling 2: Cusp 0, slope $(1, 2)$

8.2.1 Slope Transformation

- User slope (physical): $P = 1$, $Q = 2$ (cycle $1 \cdot M + 2 \cdot L$)
- Non-closable cycle: $\gamma = 0 \cdot M + 1 \cdot L$ (slope $0/1$)
- $\text{SL}(2, \mathbb{Z})$ complement: $\delta = -1 \cdot M + 0 \cdot L$
- Verification: $\det \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = 1$ (should be $+1$)
- User slope in (γ, δ) basis: $(p, q) = (2, -1)$

8.2.2 Hirzebruch–Jung Continued Fraction

$$p/q = 2/-1: [-2], \ell = 1.$$

8.2.3 g_{NZ} After Basis Change

Could not compute basis change: `apply_cusp_basis_change`: $P=0$ is even; no integer symplectic conjugate exists (need P odd)

8.2.4 Physical Weyl Vectors

$$\mathbf{a}_{\text{phys}} = (1)$$

$$\mathbf{b}_{\text{phys}} = (\frac{1}{2})$$

All hard edges are compatible ($a_j \in \mathbb{Z}$).

8.2.5 Filled Refined Index

HJ: $[-2]$, $\ell = 1$, kernel terms: 61.

$$I_{1/2}^{\text{ref, filled}} = 1 + \eta_0^{1/2} + \eta_0 + \eta_0^{3/2} + \eta_0^2 + \eta_0^{5/2} + \eta_0^3 + \eta_0^{7/2} + \eta_0^4 + \eta_0^{9/2} + \eta_0^5 + \eta_0^{11/2} + \eta_0^6 + \eta_0^{13/2} + \eta_0^7 + \eta_0^{15/2} + \eta_0^8 + \eta_0^{17/2} + \eta_0^9 + \eta_0^{19/2} + \eta_0^{10}$$

8.3 Filling 3: Cusp 0, slope (1, 2)

8.3.1 Slope Transformation

- User slope (physical): $P = 1$, $Q = 2$ (cycle $1 \cdot M + 2 \cdot L$)
- Non-closable cycle: $\gamma = 1 \cdot M + 0 \cdot L$ (slope $1/0$)
- $\text{SL}(2, \mathbb{Z})$ complement: $\delta = 0 \cdot M + 1 \cdot L$
- Verification: $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$ (should be $+1$)
- User slope in (γ, δ) basis: $(p, q) = (1, 2)$

8.3.2 Hirzebruch–Jung Continued Fraction

$p/q = 1/2$: $[1, 2]$, $\ell = 2$.

8.3.3 g_{NZ} After Basis Change

After replacing cusp basis to the NC cycle $(1, 0)$:

$$g'_{\text{NZ}} = \begin{pmatrix} 2 & 2 & 2 & 0 \\ 2 & 2 & 1 & 1 \\ \frac{1}{2} & \frac{3}{2} & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$$\nu'_x = (-2, -2), \nu'_p = (-1, 0).$$

8.3.4 Physical Weyl Vectors

$$\mathbf{a}_{\text{phys}} = (1)$$

$$\mathbf{b}_{\text{phys}} = (-\frac{1}{2})$$

All hard edges are compatible ($a_j \in \mathbb{Z}$).

8.3.5 Filled Refined Index

HJ: $[1, 2]$, $\ell = 2$, kernel terms: 1157.

$$I_{1/2}^{\text{ref, filled}} = 1 + \eta_0^{1/2} \eta^{-2V_0} + \eta_0 \eta^{-4V_0} + \eta_0^{3/2} \eta^{-6V_0} + \eta_0^2 \eta^{-8V_0} + \eta_0^{5/2} \eta^{-10V_0} + \eta_0^3 \eta^{-12V_0} + \eta_0^{7/2} \eta^{-14V_0} + \eta_0^4 \eta^{-16V_0} + \eta_0^{9/2} \eta^{-18V_0}$$